Complexity Analysis

COMPLEXITY ANALYSIS

K-WAY MERGE USING MIN HEAP

Time Complexity: O(N log K)

Space Complexity: O(K)

Maximum Heap Size: K = 3

PAIRWISE MERGING

Time Complexity: O(NK) for sequential pairwise merging in the general case.

Space Complexity: O(N) for intermediate merged results.

For this input:

N = 12 total elements

K = 3 sorted lists

Approximate pairwise comparisons:

L1 + L2: 7

Intermediate result + L3: 11

Total: 18 comparisons (using one comparison per loop iteration).

The exact comparison count can depend on the implementation's comparison-counting convention.